ASSIGNMENT 1

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1.

```
fun main() {
   println("Use the val keyword when the value doesn't change.")
    println("Use the var keyword when the value can change.")
     println("When you define a function, you define the parameters that can be passed to it. ")
      println("When you call a function, you pass arguments for the parameters.")
}
```

2

```
fun main() {
       println("New chat message from a friend")

}
```

3.

```
fun main() {
   var discountPercentage: Int = 0
   var offer: String = ""
   var item = "Google Chromecast"
   discountPercentage = 20
   offer = "Sale - Up to $discountPercentage% discount on $item! Hurry up!"

   println(offer)
}
```

4.

```
fun main() {

    val numberOfAdults = "20".toInt()
    val numberOfKids = "30".toInt()
    val total = numberOfAdults + numberOfKids
    println("The total party size is: $total")
}
```

5.

```
fun main() {

     val baseSalary = 5000
    val bonusAmount = 1000
    val totalSalary = baseSalary + bonusAmount
    println("Congratulations for your bonus! You will receive a total of $totalSalary
(additional bonus).")

}
```

6.

```
fun subtract(a: Int, b: Int): Int {
    return a - b
}

fun main() {

    val firstNumber = 10
    val secondNumber = 5
```

```
    val thirdNumber = 8

    val result = subtract(firstNumber, secondNumber)
    val anotherResult = subtract(firstNumber, thirdNumber)

    println("$firstNumber - $secondNumber = $result")
    println("$firstNumber - $thirdNumber = $anotherResult")
}
```

7.

```
/**
 * You can edit, run, and share this code.
 * play.kotlinlang.org
 */

fun main() {

    val operatingSystem = "Chrome OS"
    val emailId = "sample@gmail.com"

    println(displayAlertMessage(operatingSystem, emailId))
    println()

    val secondUserOperatingSystem = "Windows"
    val secondUserEmailId = "user_two@gmail.com"

    println(displayAlertMessage(secondUserOperatingSystem, secondUserEmailId))
    println()
```

```
    val thirdUserOperatingSystem = "Mac OS"
    val thirdUserEmailId = "user_three@gmail.com"

    println(displayAlertMessage(thirdUserOperatingSystem, thirdUserEmailId))
    println()
}



fun displayAlertMessage(operatingSystem: String, emailId: String): String {
    return "There's a new sign-in request on $operatingSystem for your Google Account $emailId."
}
```

8.

```
/**
 * You can edit, run, and share this code.
 * play.kotlinlang.org
 */

fun main() {
    val stepsTaken = 4000
    val caloriesBurned = pedometerStepsToCalories(stepsTaken);
    println("Walking $stepsTaken steps burns $caloriesBurned calories")
}

fun pedometerStepsToCalories(NumberOFStepS: Int): Double {
    val caloriesBurnedForEachStep = 0.04
    val totalCaloriesBurned = NumberOFStepS * caloriesBurnedForEachStep
    return totalCaloriesBurned
```

```
}
```

9.

```
/**
 * You can edit, run, and share this code.
 * play.kotlinlang.org
 */

fun main() {
    println(spentMoreTimeToday(300, 250))  // Output: true
    println(spentMoreTimeToday(300, 300))  // Output: false
    println(spentMoreTimeToday(200, 220))  // Output: false
}



fun spentMoreTimeToday(timeSpentToday: Int, timeSpentYesterday: Int): Boolean {
    return timeSpentToday > timeSpentYesterday
```

10.

```
/**
 * You can edit, run, and share this code.
 * play.kotlinlang.org
 */

fun printWeather(city: String, lowTemp: Int, highTemp: Int, rainChance: Int) {
    println("City: $city")
    println("Low temperature: $lowTemp, High temperature: $highTemp")
    println("Chance of rain: $rainChance%")
    println()
}
```

```
fun main() {
    // Call the function for each city with the corresponding weather details
    printWeather("Ankara", 27, 31, 82)
    printWeather("Tokyo", 32, 36, 10)
    printWeather("Cape Town", 59, 64, 2)
    printWeather("Guatemala City", 50, 55, 7)
}
```